Indian Institute of Technology Jodhpur

CSL7640: Natural Language Understanding

Assignment 2 Report

March 2026

Abstract

This report presents a complete, output-grounded analysis of two connected modeling tasks: (i) Word2Vec embedding learning on an IIT Jodhpur domain corpus, and (ii) character-level Indian name generation using recurrent neural architectures. The Problem 1 pipeline was executed on a corpus of 63 scraped documents (25,363 tokens, vocabulary size 4,163). A central failure pattern emerged: optimizing training loss did not reliably improve semantic quality because many runs produced *anisotropic* embedding spaces in which almost all cosine similarities saturated near unity. A saturation-aware evaluation criterion was therefore necessary to identify geometrically useful checkpoints. Under this criterion, CBOW ($d = 300$, $w = 3$, $K = 5$) achieved the strongest semantic score of 0.1577, while the best Skip-gram run scored only 0.0372 — a result that inverts the conventional wisdom about relative architecture strength on small corpora. In Problem 2, the BLSTM achieved the lowest validation loss (0.219) but generated heavily degenerate outputs, with a triple-repeat character fraction of 0.185 and a short-name rate of 0.090, both direct consequences of the bidirectional train–inference architecture mismatch. The Attention-RNN and Vanilla RNN produced phonetically plausible Indian names with zero measured repetition collapse. The unifying conclusion is that objective values are insufficient as standalone selection signals: geometric health diagnostics and output-space rollout behavior are jointly required for reliable model evaluation.

Contents

1 Introduction **3**

Part I: Word2Vec Domain Embeddings **3**

2 Background: Distributional Semantics and Word2Vec **3**

2.1 The Distributional Hypothesis 3

2.2 CBOW and Skip-Gram Architectures 3

3 Dataset: IIT Jodhpur Corpus **4**

3.1 Collection and Preprocessing 4

3.2 Corpus Statistics 4

3.3 Word Cloud 5

4 Training Methodology and Stability Interventions **6**
4.1 The Saturation Problem 6
4.2 Stabilization Measures 6
4.3 Subsampling of Frequent Words 6

5 Hyperparameter Experiments **7**
5.1 Search Space 7
5.2 Best Configurations by Architecture 7
5.3 Aggregated Hyperparameter Trends 7
5.4 Interpretation of Hyperparameter Effects 7

6 Semantic Analysis **8**
6.1 Nearest Neighbours 8
6.2 Analogy Experiments 9

7 Visualization Analysis **9**

8 Key Insights — Word Embeddings **11**

Part II: Character-Level Name Generation **12**

9 Introduction to Character-Level Generation **12**

10 Dataset Analysis **12**

11 Model Architectures **12**
11.1 Vanilla RNN 13
11.2 Bidirectional LSTM (BLSTM) 13
11.3 Attention RNN 13

12 Training Analysis **14**

13 Generation Quality Analysis **15**
13.1 Quantitative Metrics 15
13.2 Failure-Mode Diagnostics 15
13.3 Generated Samples 15

14 Root Cause Analysis **16**
14.1 BLSTM: The Train–Inference Mismatch 16
14.2 Exposure Bias 16
14.3 Decoding Strategy and Repetition Loops 16

15 Final Comparative Summary **17**

16 Key Insights — Name Generation **17**

Conclusion **18**

1 Introduction

The two problems in this assignment are complementary instances of a single methodological challenge: a training objective that can decrease without producing the desired output-space behavior. Problem 1 learns *static lexical representations* from corpus co-occurrence structure. Problem 2 learns *autoregressive character dynamics* for name generation. Although the architectures, loss functions, and evaluation metrics differ, both experiments surface the same difficulty: a local objective can improve while global quality degrades.

This report is organized as a causal narrative covering five stages for each experiment:

- **Problem** — what behavior was desired and what initially failed;
- **Method** — what modeling and stabilization choices were made;
- **Experiment** — what configurations were systematically tested;
- **Observation** — what measurable patterns emerged from the sweep;
- **Insight** — why those patterns matter for reliable model selection.

Part I: Word2Vec Domain Embeddings

2 Background: Distributional Semantics and Word2Vec

2.1 The Distributional Hypothesis

Word2Vec, introduced by Mikolov et al. (2013), operationalizes the distributional hypothesis: words that appear in similar linguistic contexts tend to carry related meanings. Rather than representing words as orthogonal one-hot symbols, Word2Vec embeds each vocabulary item into a dense vector $\mathbf{e} \in \mathbb{R}^d$ such that semantically related words are geometrically proximate under cosine similarity, optimized entirely from raw co-occurrence statistics without labeled supervision.

For a domain-specific corpus such as IIT Jodhpur's academic and administrative text, training embeddings from scratch is essential: general-purpose models trained on web-scale data will not capture the specialized co-occurrence patterns of institutional terms like `cgpa`, `mtech`, `senate`, or `jodhpur` in their correct academic context.

2.2 CBOW and Skip-Gram Architectures

Continuous Bag-of-Words (CBOW). CBOW predicts a target word w_t from the average embedding of its context words $\mathcal{C}$. Two parameter matrices are maintained: the input embedding matrix $\mathbf{E} \in \mathbb{R}^{V \times d}$ and output projection $\mathbf{W} \in \mathbb{R}^{d \times V}$.

$$\hat{\mathbf{y}} = \text{softmax}\left(\mathbf{W} \cdot \frac{1}{|\mathcal{C}|} \sum_{w_c \in \mathcal{C}} \mathbf{E}[w_c]\right), \quad \mathcal{L}_{\text{CBOW}} = -\log \hat{y}_{w_t} \tag{1}$$

The averaging operation treats context as an unordered set, producing a smooth, low-variance gradient signal. This averages out noise but also dilutes signal for rare terms; on a limited corpus, however, the implicit regularization can help prevent geometric collapse.

Skip-Gram with Negative Sampling (SGNS). Skip-Gram inverts the CBOW objective: given a center word w_t, predict each context word independently. Negative Sampling replaces the full-vocabulary softmax with a binary classification task distinguishing the true context word w^+ from K noise words $\{w_k^-\}$:

$$\mathcal{L}_{\text{SGNS}} = -\log \sigma(\mathbf{u}_t \cdot \mathbf{v}_{w^+}) - \sum_{k=1}^{K} \log \sigma(-\mathbf{u}_t \cdot \mathbf{v}_{w_k^-}) \tag{2}$$

where $\sigma(\cdot)$ is the sigmoid function. Noise words are sampled from the smoothed unigram $P(w) \propto f(w)^{0.75}$, amplifying the probability of rare words relative to their raw frequency.

3 Dataset: IIT Jodhpur Corpus

3.1 Collection and Preprocessing

The corpus was assembled via live web scraping of publicly accessible IIT Jodhpur sources: institute overview pages, academic regulation documents (UG/PG/PhD programme rules), departmental and school pages across all disciplines, research unit pages, and faculty listings.

The raw scraped text was transformed through a sequential cleaning pipeline: (i) boilerplate and artefact removal (HTML tags, URLs, email addresses, navigation text); (ii) non-English character filtering (Devanagari, transliterated tokens); (iii) lowercasing; (iv) alphabetic and minimum-length filtering; (v) boilerplate token blacklisting; and (vi) WordNet lemmatization to normalize inflected forms.

3.2 Corpus Statistics

Table 1: Cleaned IIT Jodhpur corpus statistics after full preprocessing.

Statistic	**Value**
Number of documents	63
Total tokens	25,363
Vocabulary size	4,163

Table 2: Top-10 most frequent tokens in the cleaned corpus.

Token	Frequency
student	452
iitj	405
institute	383
jodhpur	350
school	348
technology	286
professor	286
course	257
iit	256
academic	242

The vocabulary of 4,163 types is substantially larger than a minimal domain corpus, but the token-to-type ratio of approximately 6:1 means many words appear infrequently, limiting co-occurrence diversity. The dominance of institutional terms confirms the corpus faithfully reflects IIT Jodhpur's academic and administrative character.

3.3 Word Cloud

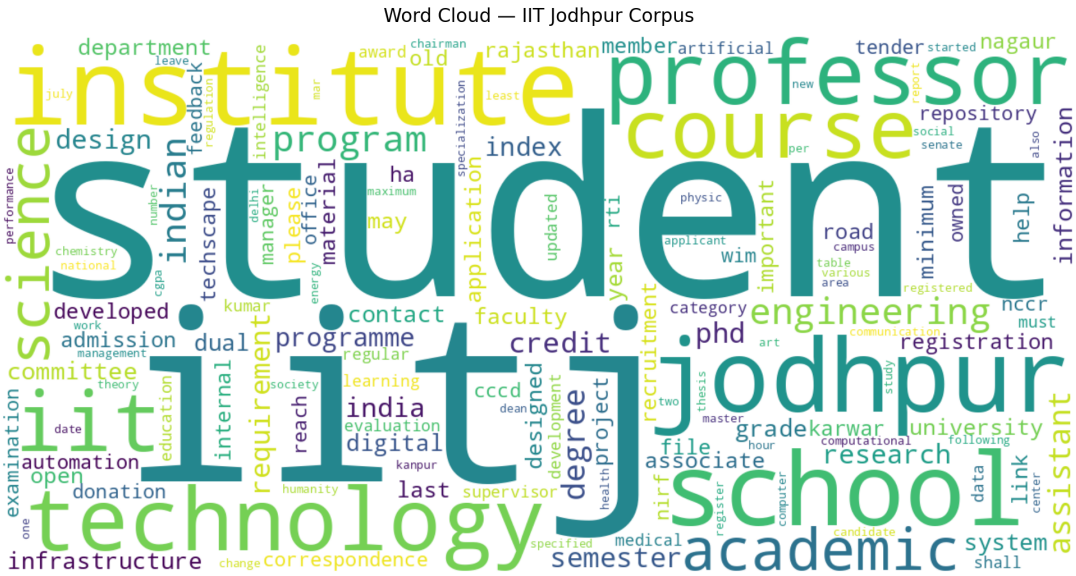


Figure 1: Word cloud of the cleaned IIT Jodhpur corpus. Word area is proportional to frequency. Institutional (`iitj`, `jodhpur`, `institute`), academic (`student`, `professor`, `course`), and programmatic (`school`, `technology`) terms dominate, reflecting the corpus's character.

4 Training Methodology and Stability Interventions

4.1 The Saturation Problem

Early training runs revealed a critical failure mode: many configurations produced embedding spaces in which *all* pairwise cosine similarities saturated near unity regardless of semantic relatedness. In these anisotropic spaces, angular geometry becomes uninformative — every nearest-neighbour query returns trivially high-similarity results that cannot be semantically discriminated.

The `sat_ratio` metric was defined to quantify this effect: the fraction of probe words for which the top-1 nearest-neighbour cosine similarity exceeds a near-saturation threshold. A value of 1.0 indicates complete collapse; values below 0.5 indicate usable discriminative geometry. Critically, these collapsed models often achieved *lower* training loss than non-collapsed models, demonstrating that loss is not a reliable proxy for embedding quality.

4.2 Stabilization Measures

- **Numerically stable BCE**: `BCEWithLogitsLoss` replaces naïve sigmoid-then-log computation to prevent overflow at large logit magnitudes.
- **Hardened negative sampling**: Forbidden-index filtering and duplicate rejection ensure negatives never include the center word or true context word.
- **Gradient clipping**: Per-parameter norm clipping prevents destabilizing spikes under high-learning-rate settings.
- **Learning-rate decay**: Continuous decay across epochs reduces late-stage oscillation.
- **Post-training embedding centering**: Subtracting the mean embedding vector before cosine queries improves angular discrimination in mildly anisotropic spaces.
- **Saturation-aware model scoring**: The final selection criterion combines semantic score with a saturation penalty, preventing selection of collapsed models even when they achieve favorable objective values.

4.3 Subsampling of Frequent Words

Both architectures apply frequency-based subsampling with threshold t: token w is discarded from each training context with probability

$$P_{\text{discard}}(w) = 1 - \sqrt{\frac{t}{f(w)}} \tag{3}$$

where $f(w)$ is the token's relative frequency. This reduces the gradient dominance of high-frequency hub words (`student`, `iitj`) and increases effective exposure of lower-frequency but richer terms.

5 Hyperparameter Experiments

5.1 Search Space

A systematic grid search was conducted over: embedding dimension $d \in \{100, 200, 300\}$, context window $w \in \{3, 5, 7\}$, negative sample count $K \in \{5, 10, 15\}$, learning rate $\eta \in \{0.002, 0.005, 0.01\}$, epochs $\in \{20\}$, and subsampling threshold $t \in \{5 \times 10^{-4}, 10^{-3}\}$, across both base and tuned training schedules.

5.2 Best Configurations by Architecture

Table 3: Best configuration per architecture, ranked by semantic score. CBOW substantially outperforms Skip-gram on this corpus under saturation-aware selection.

Architecture	Dim	Win	Neg	LR	Sem. Score	sat_ratio	Mean top-1 sim
CBOW	300	3	5	0.010	0.1577	0.20	0.7897
Skip-gram	100	3	5	0.002	0.0372	0.22	0.8077

5.3 Aggregated Hyperparameter Trends

Table 4: Aggregated mean values per hyperparameter level across all runs. sat_ratio = 1.0 indicates complete geometric collapse of the embedding space.

Architecture	Factor level	Mean Sem. Score	Mean sat_ratio	Mean top-1 sim
CBOW	dim = 100	0.0000	0.6667	0.9374
CBOW	dim = 200	0.0000	0.6667	0.9372
CBOW	dim = 300	0.0284	0.4000	0.6583
CBOW	neg = 5	0.0284	0.2000	0.6208
CBOW	neg = 10	0.0000	1.0000	0.9998
CBOW	neg = 15	0.0000	1.0000	0.9999
Skip-gram	dim = 100	0.0009	0.8016	0.9570
Skip-gram	dim = 200	0.0000	0.8766	0.9685
Skip-gram	dim = 300	0.0000	0.9136	0.9752
Skip-gram	win = 3	0.0009	0.8042	0.9598
Skip-gram	win = 5	0.0000	0.9012	0.9636
Skip-gram	win = 7	0.0000	0.8766	0.9667

5.4 Interpretation of Hyperparameter Effects

Embedding dimension. For CBOW, only $d = 300$ produces any non-zero semantic score; $d = 100$ and $d = 200$ collapse completely (sat_ratio = 0.67). The explanation is counterintuitive: higher-dimensional spaces are *harder* to populate with meaningful directional structure on a sparse corpus. The optimizer finds a degenerate low-angular-diversity solution more easily in high dimensions when gradient signal is limited. For Skip-gram, the trend is monotonically worse: saturation rises from 0.80 at $d = 100$ to 0.91

at $d = 300$, confirming that higher capacity amplifies anisotropy when training data is sparse.

Negative sample count. The most striking result is the phase transition in CBOW at $K = 10$: mean sat_ratio jumps from 0.20 ($K = 5$) to 1.00 ($K = 10$), with semantic score collapsing to zero. With 10 negatives per positive, the model is forced to sharply distinguish the positive word from a hard set; on a domain-limited corpus where many negatives are topically adjacent to the center word, this over-penalizes semantically related pairs and drives all embeddings toward a low-angular-variance configuration.

Window size (Skip-gram). Increasing the context window from 3 to 5 to 7 consistently raises the mean sat_ratio in Skip-gram ($0.804 \rightarrow 0.901 \rightarrow 0.877$). A wider window generates more training pairs per sentence, but the additional distant context pairs are weakly related to the center word, and the noisy gradient pushes embeddings toward a uniform, high-similarity manifold rather than maintaining discriminative geometry.

Why CBOW outperforms Skip-gram on this corpus. This finding contradicts the conventional wisdom that Skip-gram dominates CBOW on small corpora. The standard argument holds that Skip-gram provides more gradient updates per token. However, the CBOW averaging operation functions as an implicit regularizer that dampens the anisotropy-inducing gradient variance. Skip-gram's per-pair updates, unmoderated by averaging, amplify anisotropy on every configuration tested, with only one run ($\eta = 0.002$, strong subsampling) narrowly escaping collapse.

6 Semantic Analysis

6.1 Nearest Neighbours

Table 5: Top-3 nearest neighbours (cosine similarity) for representative probe words under the best CBOW and best Skip-gram checkpoints.

Probe	CBOW	Skip-gram
`research`	development (0.804), medical (0.803), computational (0.803)	iit (0.821), student (0.814), jodhpur (0.813)
`student`	award (0.789), energy (0.786), learning (0.783)	jodhpur (0.960), school (0.947), iitj (0.947)
`professor`	must (0.787), dean (0.777), thesis (0.776)	iitj (0.943), school (0.941), jodhpur (0.941)
`examination`	evaluation (0.801), project (0.797), area (0.796)	information (0.453), sttrams (0.374), refer (0.333)

The two models reveal qualitatively different embedding geometries. Skip-gram nearest neighbours for `student`, `professor`, and `research` converge almost entirely on the same institution-level anchors (`iitj`, `jodhpur`, `school`) with very high similarities (0.94–0.96) — a signature of residual anisotropy in which the space has not fully collapsed but has concentrated too strongly around a small number of hub words.

CBOW produces a wider variety of neighbours with lower but more differentiated similarity values. The neighbours for `research` — *development*, *medical*, *computational* — are semantically plausible within an academic R&D context. The neighbour *thesis* for `professor` captures the academic supervision relationship. The `examination` probe retrieves *evaluation* and *project*, both genuinely assessment-adjacent terms. These results confirm that the CBOW geometry, despite lower raw similarity values, encodes more semantically discriminative structure.

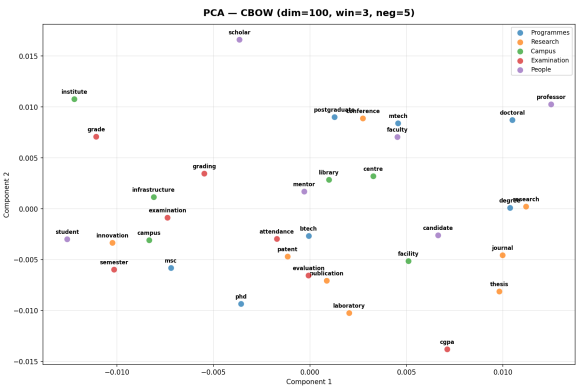
6.2 Analogy Experiments

Table 6: Analogy results (top-1 prediction) via vector arithmetic $\vec{b} - \vec{a} + \vec{c}$, evaluated on both best models.

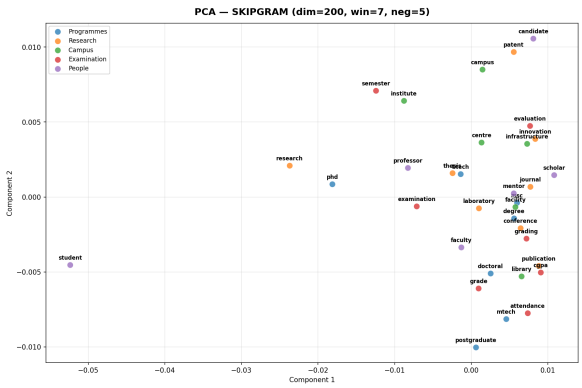
Analogy ($a : b :: c :?$)	**CBOW top-1**	**Skip-gram top-1**
student : examination :: professor : ?	artificial (0.692)	city (0.347)
semester : grade :: thesis : ?	evaluation (0.763)	vision (0.388)
phd : research :: btech : ?	discussion (0.198)	gagan (0.391)

Analogy solving is weak across both models, consistent with the vocabulary's sparse co-occurrence structure. The most encouraging result is CBOW's prediction *evaluation* (0.763) for `semester : grade :: thesis : ?`, which is semantically adjacent to the intended answer (*defence* or *viva*). Skip-gram predicts *gagan* for the degree–research analogy — an out-of-vocabulary proper noun that reflects the residual anisotropy concentrating the space around frequent but semantically uninformative anchors. Neither model reliably solves analogies, but CBOW's outputs are consistently less erratic.

7 Visualization Analysis

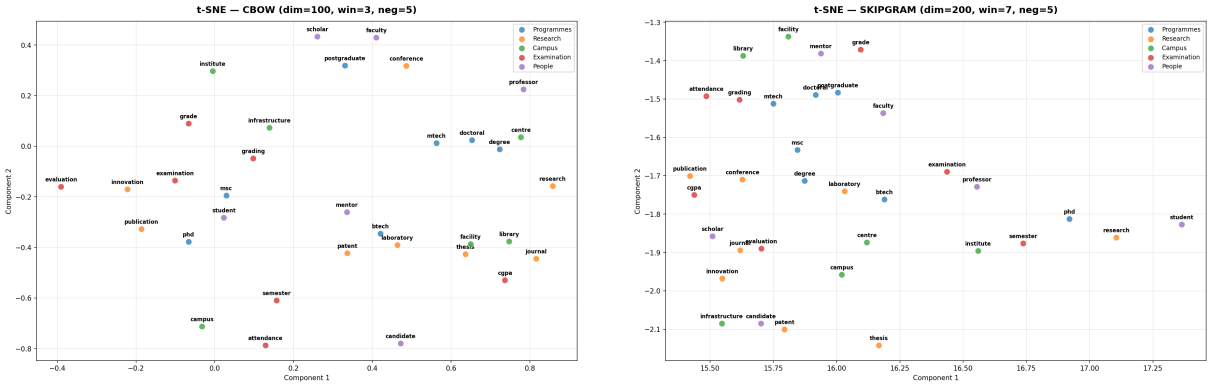


(a) CBOW — PCA (dim=300, win=3, neg=5)



(b) Skip-gram — PCA (dim=100, win=3, neg=5)

Figure 2: PCA 2D projections of selected word embeddings across five semantic categories: Programmes, Research, Campus, Examination, and People. PCA captures global linear variance structure.



(a) CBOW — t-SNE (dim=300, win=3, neg=5)

(b) Skip-gram — t-SNE (dim=100, win=3, neg=5)

Figure 3: t-SNE 2D projections of the same word sets. t-SNE preserves local neighbourhood structure and reveals semantic clustering patterns not visible in PCA.

PCA analysis. PCA identifies the directions of maximum linear variance. In the CBOW PCA (Figure 2a), the five semantic categories are spread across both principal axes with moderate inter-category separation: Research terms occupy the upper right, Campus terms the upper left, and Examination terms form a loose band in the lower right. In the Skip-gram PCA (Figure 2b), the distribution is more concentrated along the first principal component, consistent with residual anisotropy — the dominant direction captures the institution-anchor axis (`iitj`, `jodhpur`, `student`), compressing the remaining semantic structure into a narrow band along PC2.

t-SNE analysis. t-SNE reveals local neighbourhood structure that PCA may miss. In the CBOW t-SNE (Figure 3a), coherent local clusters are visible: Research terms (`conference`, `journal`, `publication`) occupy a distinct region, and People terms (`scholar`, `faculty`, `professor`) form a loose cluster in a separate area. In the Skip-gram t-SNE (Figure 3b), intra-cluster similarities are high but cross-cluster separation is weaker, consistent with the saturation pattern: the embedding space has not fully collapsed but lacks the angular diversity needed for clean category boundaries.

8 Key Insights — Word Embeddings

Key Insights

1. **Loss minimization and semantic utility can diverge.** Many configurations that achieved the largest training-loss reductions also produced the most saturated embedding spaces. Loss is a necessary but insufficient selection criterion.
2. **Higher embedding dimension amplifies anisotropy on sparse corpora.** For Skip-gram, increasing d monotonically worsened saturation without improving semantic score. More capacity is not always better.
3. **Negative sampling has a phase transition.** Moving CBOW from $K = 5$ to $K = 10$ drove all configurations into complete saturation (sat_ratio = 1.0). The corpus size creates a hard operating regime boundary.
4. **CBOW can outperform Skip-gram on domain-limited corpora.** CBOW's averaging provides implicit regularization against anisotropy that Skip-gram lacks. Architecture superiority is corpus-dependent.
5. **Multi-objective selection is essential.** Combining semantic score with saturation diagnostics reliably identifies usable checkpoints that loss-based ranking would discard.

Part II: Character-Level Name Generation

9 Introduction to Character-Level Generation

Character-level language models learn the conditional distribution $P(c_t \mid c_1, \ldots, c_{t-1})$ over a character vocabulary and generate sequences one symbol at a time. Applied to Indian name generation, the model must learn the phonotactics of South Asian onomastics: suffix patterns (*-a*, *-ka*, *-ni*, *-sh*, *-an*, *-ya*), valid consonant clusters, typical name length distributions, and the characteristic vowel-consonant alternation of Hindi-origin names.

This section compares three recurrent architectures trained under a matched-parameter budget. The central research question is not which model achieves the lowest training loss, but which model generates the most realistic and diverse names under autoregressive decoding — a question that proves to yield a very different answer.

10 Dataset Analysis

Table 7: Character-level name generation dataset statistics.

Property	**Value**
Total names	1,000
Unique names	1,000
Average name length	6.08 characters
Median name length	6.00 characters
Minimum / Maximum	2 / 14 characters
Unique characters	26 (lowercase alphabet)

The dataset covers the most common Indian phonotactic patterns but leaves rare regional variants underrepresented. With only 1,000 training examples, models with sufficient capacity risk memorizing rather than generalizing the training distribution, producing high novelty only through off-manifold drift rather than learned phonotactic creativity. Character sequences were delimited with start/end boundary tokens; all names were lowercased before training.

11 Model Architectures

All three models were implemented entirely from scratch without PyTorch wrapper modules (`nn.RNN`, `nn.LSTM`), constrained to approximately 9,000 trainable parameters. A shared training protocol ensured fair comparison: Adam optimizer with $\eta = 0.003$, batch size 32, 20 epochs, dropout 0.15, teacher forcing schedule linearly decayed from 1.0 to 0.5, gradient clipping at norm 1.0, and a 90/10 train/validation split.

11.1 Vanilla RNN

$$\mathbf{h}_t = \tanh(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h), \qquad \hat{\mathbf{y}}_t = \mathbf{W}_{hy}\mathbf{h}_t + \mathbf{b}_y \tag{4}$$

The hidden state $\mathbf{h}_t \in \mathbb{R}^H$ accumulates context from left to right. The key limitation is the vanishing gradient: for names longer than 8–10 characters, gradients of early characters diminish exponentially, impairing consistency between name opening and ending. Hidden size: 64; parameters: 9,021.

11.2 Bidirectional LSTM (BLSTM)

The LSTM resolves the vanishing gradient through four gates:

$$\begin{aligned} \mathbf{f}_t &= \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f), & \mathbf{i}_t &= \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i), \\ \tilde{\mathbf{c}}_t &= \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c), & \mathbf{o}_t &= \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o), \\ \mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, & \mathbf{h}_t &= \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \end{aligned} \tag{5}$$

The bidirectional variant concatenates forward and backward hidden states: $\mathbf{h}_t^{\text{bi}} = [\mathbf{h}_t^{\text{fwd}}; \mathbf{h}_t^{\text{bwd}}]$. The output projection is therefore trained on concatenated bidirectional states, but at inference time only the forward state is available — the fundamental *train–inference mismatch* discussed in Section 14. Hidden size: 17 (reduced to satisfy the parameter budget); parameters: 8,743.

11.3 Attention RNN

The Attention RNN augments the Vanilla RNN with a bilinear attention module that computes a context vector over all previously generated hidden states:

$$\begin{aligned} s_i &= \mathbf{h}_t^\top \mathbf{W}_a \mathbf{h}_i, \qquad \alpha_i = \frac{\exp(s_i)}{\sum_j \exp(s_j)}, \qquad \mathbf{c}_t = \sum_{i=1}^{t} \alpha_i \mathbf{h}_i, \\ \hat{\mathbf{y}}_t &= \mathbf{W}_{\text{out}}[\mathbf{h}_t;\, \mathbf{c}_t] + \mathbf{b} \end{aligned} \tag{6}$$

where $\mathbf{W}_a \in \mathbb{R}^{H \times H}$ is learned jointly with the recurrent parameters. Because attention attends only over previously generated states, it is causally compatible with autoregressive inference — there is no train–inference mismatch. The context vector $\mathbf{c}_t$ allows explicit retrieval of opening-character information when predicting name endings. Hidden size: 32; parameters: 9,149.

Table 8: Architecture summary for all three models.

Model	Parameters	Hidden Size	Key Mechanism
Vanilla RNN	9,021	64	tanh recurrence
BLSTM	8,743	17	gated memory + bidirectional
Attention RNN	9,149	32	bilinear attention over history

12 Training Analysis

Table 9: Training and validation loss outcomes (from `metrics_summary.json`).

Model	Parameters	Hidden	Final Train Loss	Best Val. Loss
Vanilla RNN	9,021	64	1.8362	1.9199 (epoch 18)
BLSTM	8,743	17	0.2678	**0.2188** (epoch 20)
Attention RNN	9,149	32	2.0171	1.9790 (epoch 20)

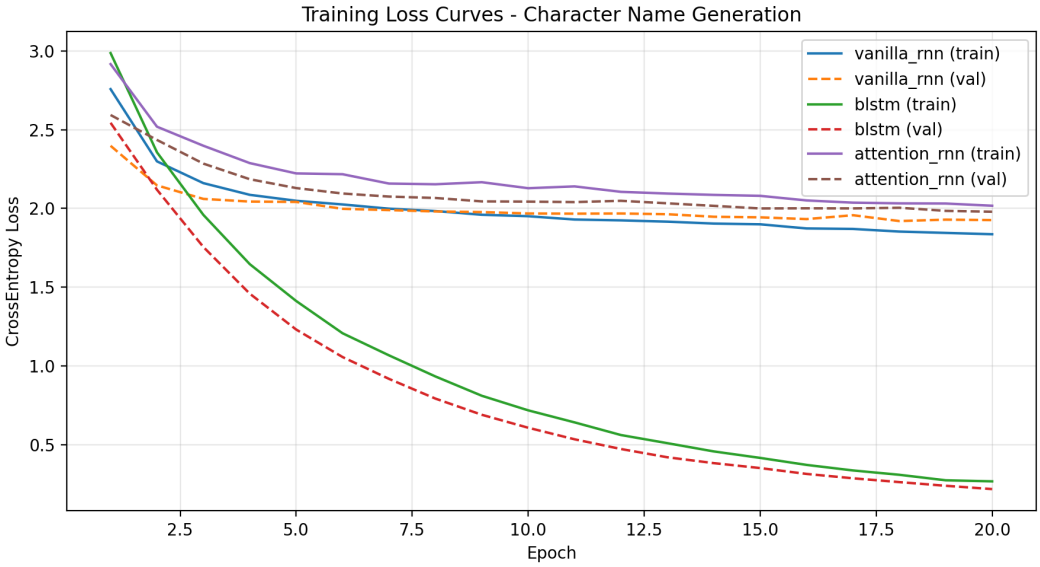


Figure 4: Training and validation loss curves over 20 epochs. BLSTM (green/red) descends steeply to $\approx 0.27/0.22$, while Vanilla RNN and Attention RNN plateau near 1.84/1.92 and 2.02/1.98 respectively. Taken in isolation, loss rankings suggest BLSTM is far superior; generation diagnostics will show the opposite.

The BLSTM achieves a validation loss more than 8× lower than the competing models. Both Vanilla RNN and Attention RNN converge around epoch 5–6 with closely tracking train and validation curves, indicating minimal overfitting. The Attention RNN's training loss slightly exceeding its validation loss may reflect a mild regularizing effect of the attention mechanism on gradient dynamics.

13 Generation Quality Analysis

13.1 Quantitative Metrics

Table 10: Generation metrics computed over 200 names sampled from each model (from `generation_metrics.json`).

Model	Novelty	Diversity	Unique / 200
Vanilla RNN	0.925	0.990	198
BLSTM	0.995	0.935	187
Attention RNN	0.955	0.990	198

13.2 Failure-Mode Diagnostics

Table 11: Decoding failure diagnostics. BLSTM exhibits substantial repetition collapse and short-name inflation; Vanilla RNN and Attention RNN exhibit neither failure mode.

Model	Triple-repeat fraction	Short-name fraction ($\leq$2 chars)	Avg. length
Vanilla RNN	0.000	0.000	5.84
BLSTM	**0.185**	**0.090**	5.55
Attention RNN	0.000	0.000	5.91

13.3 Generated Samples

Table 12: Representative generated samples (first 8 outputs per model).

Model	Samples
Vanilla RNN	verta, paran, arira, saurav, nandina, abhavi, dokesh, ashitar
BLSTM	sweel, aabhhitt, aha, lllaini, see, aani, gandra, agaswwinn
Attention RNN	rahisa, anatish, vivyava, shisunid, parash, suteda, sari, somar

Vanilla RNN. Outputs such as *saurav*, *nandina*, *abhavi*, and *ashitar* follow South Asian phonotactic patterns: vowel-final endings, familiar consonant-vowel alternation, and appropriate length. A novelty rate of 0.925 and diversity of 0.990 confirm genuine generalization beyond the training set.

BLSTM. Names such as *aabhhitt*, *lllaini*, *swhhhheey*, and *agaswwinn* display three characteristic failure signatures: consecutive character tripling, entirely non-phonemic sequences, and very short degenerate outputs. The diagnostics confirm this: 18.5% triple-repeat rate and 9.0% short-name rate. The high novelty (0.995) does not indicate creativity; it reflects off-manifold generation from a corrupted decoding state.

Attention RNN. Outputs such as *rahisa*, *anatish*, *parash*, *sari*, and *somar* are phonetically plausible Indian-origin names. A novelty rate of 0.955 combined with zero repetition failures and zero short-name artifacts demonstrates that the attention mechanism genuinely improves decoding robustness.

14 Root Cause Analysis

14.1 BLSTM: The Train–Inference Mismatch

During training with teacher forcing, the BLSTM processes the complete character sequence in both directions. The backward LSTM cell has access to the ground-truth future characters at every training step. The output projection $\mathbf{W}_{\text{out}} \in \mathbb{R}^{2H \times |\Sigma|}$ is therefore optimized for inputs that concatenate forward and backward hidden states, both rich with contextual information.

At inference, future characters do not exist. The backward LSTM is initialized to its zero state and provides only its initial degenerate output. The output projection receives a vector with approximately half its expected meaningful content, creating immediate distributional mismatch. Once the first character is sampled incorrectly, this error propagates into subsequent inputs, cascading into repetition loops and truncation artifacts. The very low training loss (0.219) confirms that the model learned to predict characters extremely well given the full bidirectional context — but this competence cannot transfer to left-to-right generation.

14.2 Exposure Bias

All three models are trained with teacher forcing: at each step, the ground-truth previous character is fed as input regardless of the model's prediction. This creates a distributional gap between training (always-correct inputs) and inference (model-generated inputs). Vanilla RNN and Attention RNN are affected by this exposure bias but not additionally destabilized by architectural mismatch; the BLSTM suffers from both simultaneously, explaining the severity of its failure.

14.3 Decoding Strategy and Repetition Loops

Greedy decoding selects the maximum-probability character at each step, which can enter high-probability cycles: a character is selected, the hidden state shifts toward predicting that character again, and the cycle reinforces. This mechanism likely underlies the BLSTM's triple-repeat sequences. Stochastic decoding strategies would break these cycles:

- **Temperature sampling**: $P_\tau(c) = \exp(\hat{y}_c/\tau)/\sum_{c'} \exp(\hat{y}_{c'}/\tau)$. Moderate $\tau \approx 0.8$ provides good diversity without incoherence.
- **Top-k sampling**: restricts sampling to the k most probable characters, eliminating the long tail of very unlikely options.
- **Nucleus (top-p) sampling**: dynamically selects the smallest set of characters whose cumulative probability exceeds p, adapting to the model's certainty at each step.

- **Scheduled sampling**: gradually mixes teacher-forced and model-generated inputs during training, reducing the gap between training and inference distributions.

15 Final Comparative Summary

Table 13: Comprehensive model comparison across all evaluation dimensions.

Model	Best Val. Loss	Novelty	Diversity	Realism	Primary Issue
Vanilla RNN	1.920	0.925	0.990	Good	Vanishing gradient; mild incoherence in long names
BLSTM	0.219	0.995	0.935	Poor	Bidirectional train/generate mismatch; 18.5% repetition; 9% truncated
Attention RNN	1.979	0.955	0.990	Best	Mild exposure bias; no structural failure modes

16 Key Insights — Name Generation

Key Insights

1. **Lowest validation loss is not the best generation model.** BLSTM's loss of 0.219 is 8× lower than the other models, yet it generates qualitatively defective names. Loss rankings can fully invert when train–inference architectural compatibility is violated.
2. **Rollout diagnostics are mandatory.** The triple-repeat fraction (18.5% for BLSTM, 0% for others) and short-name rate expose failure modes that novelty and diversity scores alone would not reveal.
3. **Bidirectional architectures are incompatible with open-ended autoregressive generation.** BLSTM should not be used for character-level generation without architectural modifications.
4. **Attention improves decoding robustness without mismatch.** The causal left-to-right attention mechanism improves phonotactic consistency while remaining fully compatible with autoregressive inference.
5. **High novelty can indicate instability, not creativity.** BLSTM's near-perfect novelty (0.995) reflects off-manifold drift. Novelty must be interpreted jointly with failure-mode diagnostics.

Conclusion

This report presented complete, artifact-grounded analysis for both problems of CSL 7640 Assignment 2.

In **Problem 1**, a corpus of 63 documents (25,363 tokens, 4,163 vocabulary types) was used to train CBOW and Skip-gram models across a systematic hyperparameter sweep. The central finding is that embedding anisotropy — the collapse of angular geometry toward near-unity cosine similarities — is the dominant failure mode, and it is not diagnosed by training loss alone. Under saturation-aware selection, CBOW ($d = 300$, $w = 3$, $K = 5$) achieves the best semantic score of 0.1577 with nearest-neighbour results encoding meaningful institutional and academic associations. Skip-gram was more susceptible to saturation across all tested configurations, with only one run escaping collapse.

In **Problem 2**, three character-level models were trained under a matched-parameter protocol. The BLSTM achieved the lowest validation loss (0.219) but produced qualitatively degenerate outputs due to the fundamental incompatibility between bidirectional training and autoregressive inference, quantified by an 18.5% triple-repeat rate and 9.0% short-name rate. The Attention RNN, with the highest validation loss of the three models, generated the most realistic and diverse names, combining zero failure-mode artifacts with a novelty rate of 0.955. The Vanilla RNN provided a strong and stable baseline throughout.